

Participation Report at the Biennial Africa Climate-Smart Agriculture Conference (FARA) in Ghana

Joab Osumba

Conference Participation Report



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**Accelerating Impacts of CGIAR Climate Research for Africa
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About AICCRA

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1 Background information

1.1 Introduction

Many attempts have been made at the global level to tackle the issues of climate adaptation and resilience of food systems and agriculture. Attention has been focused on the concept and practice of climate-smart agriculture (CSA), which seems to be well established as a keyword in agricultural development. The Forum for Agricultural Research in Africa (FARA) and other organizations that are coordinating agricultural research in Africa are channeling attention to the issues of CSA, through scoping studies and mapping of various CSA initiatives to identify the state of knowledge, available technologies, and institutional arrangement to advance African agricultural development. Since 2020 FARA and partners organise a biennial CSA conference as a forum for discussion among the different stakeholders involved in the generation and use of climate-relevant agricultural knowledge, technologies, and innovations in Africa. The conference forms a major platform for stock-taking, reporting on initiatives, and joint identification and prioritization of the next line of action, especially in technology generation and use. The conference also engages in policy and advocacy discussions, and includes presentations of research papers, to nurture the policy and technical dialogue among academicians, researchers, managers, policymakers, small producer organizations, civil society organizations, agribusiness sector, and students.

1.2 Overview and background information

The urgent need to mobilize science and innovation for the benefit of climate-smart agricultural development in Africa cannot be overemphasized. The general view in Africa emphasizes the importance of implementing policies to promote climate-smart agriculture, supported by investments and backed by high-quality scientific research and innovation. Accelerating Impacts of CGIAR Climate Research for Africa (AICCRA) fills a critical gap in helping to make cutting-edge CGIAR research and innovation available to National Agricultural Research Systems (NARS) and other key stakeholders in Africa. AICCRA not only helps take to scale the most strategic and impactful CGIAR-CCAFS-Africa programs, promoting resilience to climate change and improved food security in the region but also supports critical knowledge creation and capacity building activities to enable regional and national-level stakeholders to take CSA innovations to scale. Further, AICCRA fosters partnerships between CGIAR and local research institutes, universities, civil society organizations, farmer organizations, and private sector, and promotes the adoption of CSA solutions across sub-regions within Africa (concentrated in six anchor countries: Ghana, Mali, Senegal, Kenya, Ethiopia, and Zambia). In these countries and regions AICCRA supports the scaling out of CGIAR research-based CSA innovations through regional and national partners.

1.3 Biennial Africa Climate-Smart Agriculture (CSA) Stakeholders Conference

The 2022 edition of the Biennial Africa Climate-Smart Agriculture (CSA) Stakeholders Conference was held at Alisa Hotel in Accra, Ghana, from 14th to 16th of September 2022. The conference gave an opportunity to take stewardship of the CSA actions especially the spate of technology generation and a foresight analysis of the plausible

futures. The conference also hosted the continental dialogue on the Africa Climate-Smart Agriculture Framework (ACSAF). ACSAF is developed to respond to and plan the implementation of the AU-developed Africa Climate Change Strategy. The ACSAF dialogue also provides the opportunity to prepare adequately for CoP27 in Sharm El-Sheikh, Egypt. The conference also hosted side events for 1) stakeholders' consultation towards the development of the Soil Initiative for Africa (SIA), and 2) the meeting of the High-Level Technical Think-Tank set out to respond to emerging technological issues in African agriculture. The hybrid (physical and virtual) event, hosted by FARA and its key partners, provided opportunities for researchers, academicians and students to share their research findings, and for policymakers, technocrats, private sector actors, and members of the civil society organisations to have dialogue on pertinent issues in managing climate change in African agriculture. The conference provided an international platform for engagement on dealing with climate change to boost food production in Africa.

Apart from the 1) 2022 Biennial Africa CSA Stakeholders Conference the FARA event, dubbed "Science and Partnerships for Agriculture Conference (SPAC 2022)", included three other parallel events namely 2) Knowledge Management for Agricultural Development (KM4AgD), 3) Pan-African Network for Economic Analysis of Policies (PANAP), and 4) Long-Term Europe-Africa Research and Innovation Partnership for Food and Nutrition Security and Sustainable Agriculture (LEAP4FNSSA).



SPAC serves as the platform for key actors to update the continent's preparedness to cope with the impacts of climate change in African agriculture as well as contribute to the delivery of continental instruments like Agenda 2063.

1.4 Conference Objective and Theme

The conference sought to address pertinent issues affecting agricultural productivity on the continent, to strengthen research and to develop action on topical issues in African agriculture to ensure productivity and competitiveness in the sector that engages a larger proportion of the working population on the continent. It also served as a platform for key actors to update the continent's preparedness to cope with and mitigate the impacts of climate change in African agriculture as well as contribute to the delivery of continental instruments like Agenda 2063. It assembled actors in including policymakers, farmers, technocrats, researchers, and the private sector to facilitate the introduction of the much-needed coordination effort for climate change actions at the continental level.

Conference Theme: The main theme of the conference was "Introspection on CSA actions to strengthen accountability, resource use and impact in Africa". Six exciting sub-themes (topics) were deliberated in various sessions to bring various perspectives and practices, viz: -

1. The role of digitalization in advancing CSA in the smallholder systems

2. The nexus of CSA and the mechanization of the smallholder system in Africa
3. Bottom-up and system-wide capacity development approaches to enhance CSA practices
4. Strengthening the support of extension and advisory services to ensure CSA compliance among smallholders in Africa
5. Advances in CSA technology generation and use in crop, livestock, fisheries and aquaculture
6. The contribution of agroecology to CSA

1.5 Conference Methodology

The Africa Stakeholders Conference on Climate Smart Agriculture is facilitated by the Forum for Agricultural Research in Africa (FARA), European Union, and other partners. The conference is co-organized by the Member organizations of the CAADP-XP4 in collaboration with several partners, donor agencies, authorizing agencies and Government of Ghana. The Conference was fashioned to capture broad-based partnerships at different levels to address key issues in Africa's agricultural research and innovation space. The four-in-one conferences were convened concurrently, each running its programs and a series of side events separately but with joint opening and closing plenaries. The joint conference provided an opportunity to leverage the participation of high-level individuals in African and European agriculture to jointly address pertinent emerging issues affecting the continent. The four conferences and their constituents provided important technical and policy information from different engagements.

1.6 Expected outputs of the workshop

The outputs of this Africa Stakeholders Conference on Climate Smart Agriculture were expected to contribute to the Africa Position at the next Conference of Parties on Climate Change (CoP27) in Sharm El-Sheikh Egypt in November 2022.

Session 1: Opening Session and setting the stage

The joint opening session <https://www.youtube.com/watch?v=9Di1l4b9zw4> was graced by the following dignitaries: -

- Opening Remark Dr. Alioune Fall Chair of FARA
- Goodwill message Dr. Hans Hoogeveen GACSA co-chair
- Goodwill message ADG- FAO Dr Abebe Haile Gabriel
- Video message Stephen Quest DG JRC European Union Commission
- Mariya Gabriel, European Commissioner for Innovation, Research, Culture, Education and Youth
- Goodwill message Massimo Mina, Head of Cooperation, Delegation of the European Union to Ghana
- Goodwill Video message HE Nardos Bekele Thomas CEO of AUDA-NEPAD
- Video message HE Josefa Sacko Commissioner of DERBE, AUC

- Video message Dr. Akin Adesina President of Africa Development Bank
- Welcome Address Dr. Owusu Afriyie Akoto Minister of Food and Agriculture, Ghana

The opening speeches were followed by a Keynote Address by HE. Jeanine Milly Cooper Minister of Agriculture, Liberia, and a Lead Paper on “Introspection on CSA Action to Strengthen Accountability, Resource Use and Impact in Africa” by Dr. Bruce Campbell (ClimEAT). Thematic keynote papers for the CSA sessions of the Conference were presented by the following participants: -

1. The role of digitalization in advancing CSA in the smallholders’ systems Facilitator: Dr. Vincent Aduramigba [GCA]
2. The nexus of CSA and the mechanization of the smallholder system in Africa. Facilitator: Dr Mkomwa [ACTN]
3. Bottom-up and systemwide capacity development approaches; to enhance CSA practices Facilitator: Prof. Babalola
4. Strengthening the support of extension and advisory services to ensure CSA compliance among smallholder farmers in Africa Facilitator: Max Olupot [AFAAS]
5. Advances in CSA technology generation and use in the crop, livestock, fisheries, and aquaculture. Facilitator: Dr Alcade Segnon [AICCRA]
6. Compatibility assessment of agroecology and CSA practices Facilitator: Dr. Cliff Dlamini [DeSIRA-Lift]

Day 1 ended with an ACSAF Plenary

https://faraafrica.zoom.us/rec/share/9zOzhcj2uwQR1mL7QQPJaklhKeAhqyf7fXo22En_HL8wbSKYE9B3CGRsaZgeh5a.MedXXfiYG2uqrx7b

Session 2: Sub-Thematic Area 3 “Bottom-up and system-wide capacity development approaches to enhance CSA practices”

Sub-theme 3 was focused on “**Bottom-up and system-wide capacity development approaches to enhance CSA practices**”.

2.1 Context of the topic (background to the presentation)

CSA has the potential bottom-up and system-wide capacity to deliver “triple wins” by contributing to sustainably increasing productivity and food security; enhancing farmers’ adaptive capacity and reducing greenhouse gas emissions. Despite the potential of climate-smart agriculture in promoting bottom-up and system-wide capacity development, there is the dearth of workable approaches for enhancing climate-smart agriculture practices. The decision on whether or not to adopt a particular climate-smart agriculture practice is influenced by many idiosyncratic and covariate factors. These factors include the farmer’s awareness of the CSA practice, the amount and quality of information at the farmer’s disposal, the

financial (dis)incentives to adopt, the farmer's time preference, and the level of risk averseness. Overall, the desired scaling-up of climate-smart agriculture includes identifying specific practices, knowledge, and technologies within conducive enabling environments (social and economic) and with institutional and policy arrangements. The scaling-up process for CSA may occur horizontally by replicating promising/ proven practices and technologies in new geographic areas or target groups or vertically by catalysing institutional and policy change or diagonally by adaptive management within project implementation to reflect emerging reality. The papers scheduled for presentation in this sub-theme on bottom-up and system-wide capacity development approaches to enhance climate-smart agriculture practices are classified into the following areas: -

1. Biofortification, yields, and food systems
2. Climate-smart agriculture practices by smallholders' systems
3. Agricultural value chains and financial services
4. multi-stakeholder platforms for promoting climate-smart agriculture
5. Empowerment, entrepreneurship, and emerging markets

Thematic Paper on **“a replicable innovative tool for adaptive climate monitoring and weather forecasting using traditional indigenous and local indicators to strengthen agro-weather resilience at scale”** was presented on 14th September 2022.

2.2 Background to the tool

In the absence of reliable conventional weather information, many farmers and pastoralists resort to indigenous climate knowledge to fill the gaps, using traditional and local indicators. Such indicators may include biological (behaviour of plants, animals), ecological (behaviour of ecosystems, etc.), physical (behaviour of objects and materials, position and motion of objects, water-related features e.g., dew, etc.), celestial or astrological features (behaviour of whirlwinds, moon, etc.), behaviour of the human body (sensations), etc., all of which are validated through observation and monitoring. It is often argued that the problem with scaling traditional indigenous or local knowledge (ITK/ LTK) of climate and weather is because it is context-specific and is not often documented, mainly sustained from one generation to another through oral history and local expertise or skills, and that the elders who are the custodians of this knowledge, institutional memory, skills and experience are exiting the scene, creating a wide inter-generational gap between the elders and the current generation. It is also often argued that these challenges are likely to worsen with climate change, mainly due to the changing behaviour and/or progressive disappearance of traditional indicators of local weather monitoring to determine a forecast, a situation which, it is argued, threatens the ability of crop producers and pastoral livestock keepers to cope with and adapt to climate change [28]. Often there is pressure on practitioners, researchers and stakeholders to collect, preserve, validate and adopt these ITK/ LTK indicators. However, the initiatives introduced so far to address this problem have been largely project-based, *ad hoc* and

academic, with no clear decision-support tool to move the conversation to a practical and sustainable model, always fizzling out as soon as the project closes.

Currently there are weak mechanisms for collecting information on climate risks to help tailor the subsequent content for proper integration. What often happens in such pilots is merely to ask subjective questions about whether traditional weather predictions correlate with conventional forecasts, without any corresponding empirical comparison [15]. What is basically needed in the changing circumstances is not necessarily the uniformity of indicators or the presence historical experts or mystical or ritualistic rainmakers but a robust, easy to administer decision-support tool that can be owned by local communities and be applicable across different ethnographic and socio-cultural contexts. Instead of looking back to mystical or ritualistic rainmakers (all male) who are often secretive and not sustainable, or elders (all male) who are increasingly losing their grip on the traditional knowledge and institutional memory due to climate change and/or leaving the scene by natural attrition, making it irreplicable, unscalable and impossible to institutionalize [11], [29]–[31], it is possible to build a new set of knowledge using a new tool by opening the conversation with the whole community (both male and female, both elderly and youthful) to identify emerging indicators and experiences [32]. This approach can support empowerment of women and youth through recognizing and valuing their knowledge and roles as change agents in using ITK/ LTK and can help to ensure that valuable ITK/ LTK held by local communities' women is not missed out in efforts to integrate scientific knowledge and ITK/ LTK of climate solutions. It is against this background that an innovative decision-support tool was conceptualised and developed to bridge the gaps to meet users' felt needs.

2.3 The TILKIT tool

This paper presents a replicable innovative decision support tool – herein called the traditional indigenous and local knowledge tool (TILKIT) – to systematize ITK/ LTK base for local climate monitoring and weather forecasting. The tool was conceptualized, developed, tested and validated in various local level training sessions under two training-of-trainers (ToT) initiatives and farmer group trainings on Climate-Smart Agriculture (CSA) in East Africa from March 2016 to December 2021. Through the new tool, ITK/ LTK indicators that have become increasingly less reliable can be recalibrated and validated. The tool uses consensus-based ethnographic indicators of local climate and weather to overcome the challenges of local climate and weather information to satisfy users' felt needs. It is a versatile Community-based Adaptation (CbA) decision-support tool for local climate monitoring and weather forecasting developed for effective validation and scaling of traditional climate monitoring and weather forecasting in various localities. The tool adopts a transdisciplinary or participatory learning and action research (PLAR) model to drive self-organization processes and to bring ITK/ LTK of climate into the common knowledge pool in each community. It is meant to build local capacity for local weather forecasts and agro-weather advisories using consensus-based ethnographic weather indicators to improve climate extension service delivery.

2.4 Methodology for the tool

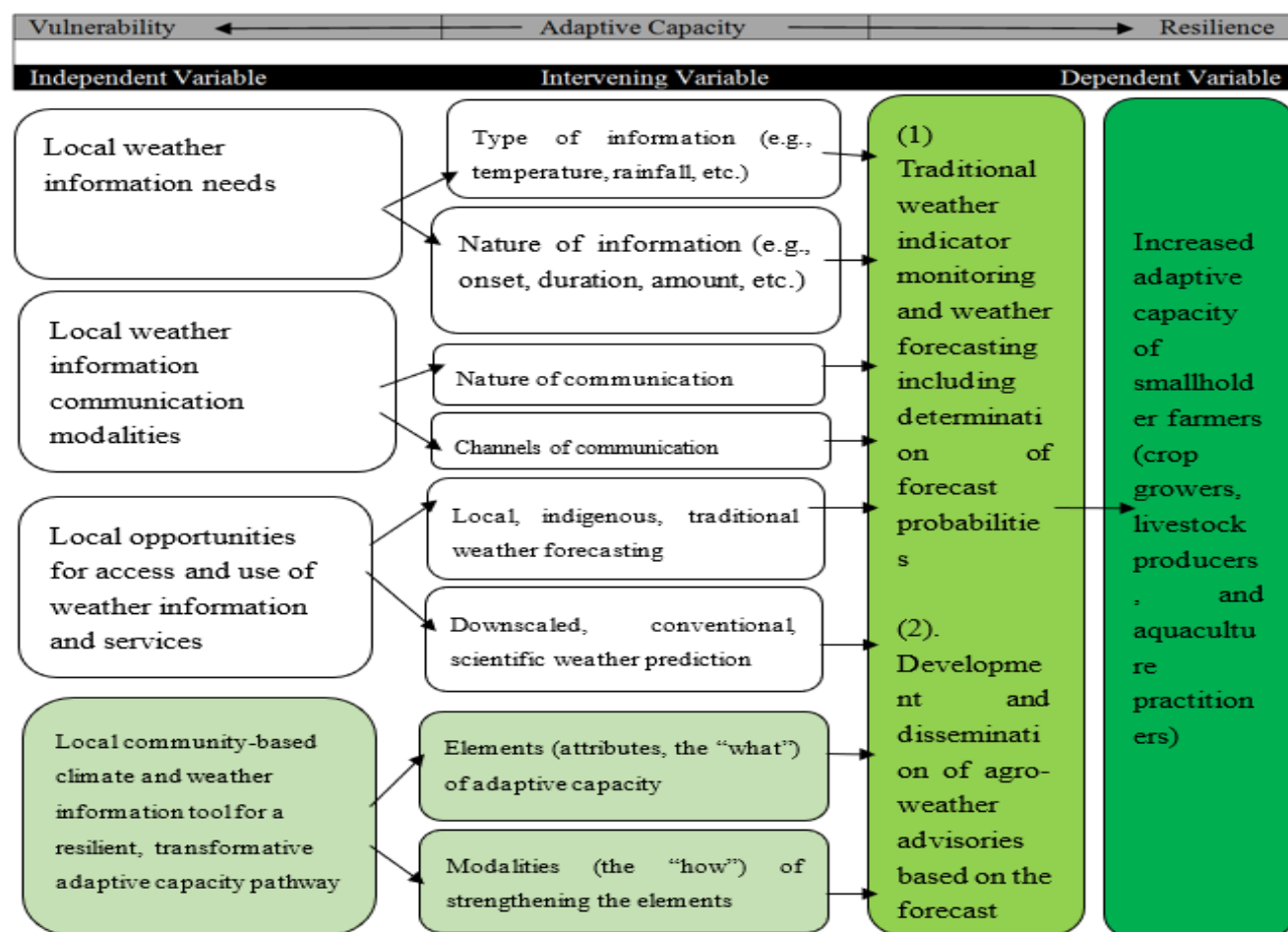
The study began with a desktop synthesis of relevant information. Fieldwork was conducted in the form of ToT and FGD with community group leaders. Participants were divided into discussion groups to identify locally observable indicators, describe what the indicators mean (=the likely scenario), and suggest advisories (=possible action required) based on the indicator and its meaning for local climate and weather forecasting. The developed scenarios and advisories (using agreed indicators) were presented in plenary, discussed to assess their validity and usefulness. The approach is to 1) identify (=groups propose in groupwork) and confirm (i.e., plenary concurs in discussion) known local indicators that signal rain issues, 2) describe what that indicator means in terms rainfall forecasting in the locality, 3) determine whether or not the indicator has been observed in the season, 4) where it has been observed (if already observed), and 5) interpret what the observation means for the coming season – whether based on the indicator the rains can be normal, above normal or below normal, where normal is defined as local historical expectation. Depending on the number of indicators the community identifies and the number of signs each indicator exhibits, it is possible to tally the observations and estimate percentages of the normal, above or below normal rainfall predictions in the community, with clear lead times, and derive locally appropriate advisories for the season from the individual indicator advisories. The recorded scenarios and advisories were merged, tallied, analysed, integrated, packaged and reported to determine local weather forecast probabilities.

2.5 Conceptual Framework

The conceptual framework for the TILKIT tool is based on the variables that feed into the innovative tool for indigenous weather forecasting (Figure 1). The concept comprises four key elements, viz.,

- knowledge of local climate risks and local vulnerabilities
- monitoring of local weather using local indicators,
- analysis and forecasting of weather hazards based on indicator behaviour,
- dissemination and communication of advisories (warnings, alerts and suggestions) based on the forecasts; and
- local capabilities to respond to the warnings generated and/or received

Figure 1: Conceptual Framework for the TILKIT tool



2.6 Participation landscape for TILKIT tool development

Testing and validation were conducted with 1,127 participants among various communities across Kenya, Tanzania and Uganda. The participation landscape for development of the tool is shown in Table 1.

Table 1: Participation landscape for TILKIT tool development

Country	Cohort	Date	Total # of participants	Remarks
Kenya	Homa Bay and Migori	March 2016 and February 2017	129	STARCK+ Farmer representatives from one sub county in Migori and six sub counties in Homa Bay during CSA trainings
	Siaya	March 2016 and March 2017	150	STARCK+ Farmer representatives from five sub counties in Siaya during CSA trainings
	Kisumu	March 2017	67	STARCK+ Farmer representatives from six sub counties in Kisumu during CSA trainings
	Kisumu	March 2019	33	CRAFT and AICCRA – representing two farmer aggregators reviewing for CRAFT the work done by CCAFS from 2011 to 2018
	Kitui and Nakuru	August – September 2019	107	CRAFT and AICCRA – representing ten farmer aggregators and one local service provider
	Nakuru, Kisumu, Machakos, Narok, Embu,	June– July 2021	87	CRAFT and AICCRA – representing Farm to Market Alliance or FtMA [Cereal Growers Association (CGA) and Participatory Approaches for Integrated Development (PAFID)] and County Governments of Nakuru, Narok, Machakos and Embu, in a collaboration between CRAFT and the FtMA Initiative implemented in Kenya by World Food Programme (WFP)
Tanzania	Dodoma, Singida and other venues	September – December 2019	215	CRAFT – representing nine farmer aggregators
Uganda	Lira, Gulu and other venues	July – December 2019	339	CRAFT and AICCRA – representing 35 farmer aggregators
Total			1,127	Three countries; 6 years

2.7 Key findings

Results of the TILKIT tool comprise identified indicators per locality, and their implications for local weather forecasting, which for the first time is presented in a systematic manner, in probabilistic terms that local communities can associate with, and which can compare empirically with conventional weather forecast language. A screenshot image of the tool is presented in Figure 2, using actual groupwork sample. A sample of identified indicators are presented in Table 2. From Table 2, it is shown that the shortest lead time in the forecast is one day, and the longest lead time is three months.

Figure 2: Example of actual groupwork output using the tool, Narok County, Kenya, July 2021

Groupwork: Traditional, indigenous and local community weather forecasting exercise:

In groups, share your experiences on traditional indigenous and local knowledge indicators (TILKI) of rainfall in your community

A1	B2	C3	D4	E5		
Traditional, indigenous and local indicator of weather in your area	Local meaning for weather in the locality (in terms of whether there will be rain or drought, or whether the rains will be normal, more or less)	How long does it take for the prediction after witnessing the indicator	Have you already seen it this season? – meaning do we have a forecast based on this indicator	In view of, and based on, the information so far available from the weather indicator, by the rating of this locality:		
				Is the rain for the season likely 0) no forecast yet 1) above normal or 2) normal, 3) below normal?	When (dates range) is the rain likely to start or stop if it will happen?	What farming plans or actions, or strategies will you adopt based on this traditional weather forecast information available to you?
Dew after long dry spell	Onset of rain	1 week	yes	2	Rains expected within days	Planting, planning for fertilizer application.
Clouds at 5am	Onset of rain	1 week	yes	2	Rains expected within a week	Planting, weed control
Lightning at the west	Onset of rain	1 week	no	0	Days after occurrence	Planting, weeding, fertilizer application.
Very Chilly morning	Beginning of dry spell	2 days	no	0	After a long time (not defined)	Water conservation, mulching,
When Moon disappears	Onset of 1 week-long rain	3 days	Yes	3	Days after occurrence	Planting, setting of farm structures.
Total tally for normal, above normal and below normal scores	Summary of meanings for the indicators	Summary of lead times for indicator prediction	Tallies for Yes and No	Above normal = Y% Normal = X% Below normal = Z%	Summary of expected dates of onset	Summary of adaptation options suggested

Table 2: Demonstration of the TILKIT weather forecast using traditional and local indicators

Indicator category	A1	B2	C3	D4	E5
	Traditional indicator of indigenous weather prediction in the community/ locality/ area	Description: Local meaning for weather – Local meaning for weather in the locality (In terms of whether there will be rain or drought, or whether the rains will be normal, more or less)	How long does it take for the prediction, after witnessing the indicator?	Forecast (Y=Yes)	In view of, and based on, the information so far available from the weather indicator, by the rating of this locality: Is the rain for the season likely: 1) above normal or 2) normal, 3) below normal?
Celestial/ Astrological	Whirling wind from the lake	Depressed rain within a week	1-2 weeks	Y	Below Normal
	Whirlwind-to east	Rainfall	One week	Y	Above normal
	Whirlwind-West wards	Dry spell	One week	Y	Below normal
	When at Mwingi and the new moon faces Tanzania	Dry spell	3 months	Y	Below normal
	Full moon with halo (kiyuuviyo)	Rain expected	1-2 weeks	Y	Above normal
Physical	Absence of dew on leaves in the morning	Rain on the same day	1 day	Y	Normal
	Morning Dew & fog	Dry spell	1-2 weeks	Y	Below Normal
	Lightning around Kisumu and Tindereet area	Rainfall	2/3 months	Y	Above normal
Biological	Regeneration of new leaves in acacia tree species/ Acacia starts to sprout	rains are approaching	1-2 weeks	Y	Normal to above normal
	Blooming of baobab trees.	Rains in few weeks	2-3 weeks	Y	Normal
	Shedding of leaves- Baobab	Dry spell	1-2 weeks	Y	Below normal
	Frequent encounter of snakes, rodents and rabbits etc out of their habits	Dry spell	One month	Y	Below normal
	Touching down of dragon flies	Rain same day	1 day	Y	Normal to above normal (when they are many) Below normal (when they are few)
	Livestock restlessness	Dry spell	2 weeks	Y	Below normal
	Livestock oversleeping	More rain	2 weeks	Y	Above normal
	Croaking of frogs	Rain expected	1 to 2 weeks	Y	Normal
	Bee migration	Dry spell	1 to 2 weeks	Y	Below normal
	White butterflies	Rainfall	After-two weeks	Y	Above normal
	White birds/ chebiswet	Rainfall	One day	Y	Above normal
	Army worm	Dry spell	2 months	Y	Below normal
	Presence of red ants/ kumbe kumbe/ kongaeek	Rain	One week	Y	Above normal
Human body behaviour (sensing)	High temperature (Yuuau)	Rain	1 to 2 weeks	Y	Above normal
	Soil smell- "pleasant"	Rain	1-2 weeks	Y	Normal
Forecast for the season: Total tally for normal, above normal and below normal scores; Summary of meanings for the indicators; Summary of lead times for indicator prediction			Above normal = 8 Normal to above normal = 6 Below normal = 10 Total = 24	Y	Above normal = 33% Normal to above normal = 25% Below normal = 42%

As shown in Table 3, TILKIT indicators with long lead times can be used for strategic planning while those with short lead times can be used for tactical and/or emergency decision making [37], [38]. A lead time of two-to-three months can be used for strategic seasonal planning by the community. Lead times of one week to one month can be used for tactical and operations decisions in the community and on the farm. One-to-two-day or less than one week lead times lead times can be used for emergency decisions on-farm.

Table 3: strategic, tactical, operational and emergency phases of planning and decision making

Response level Forecast level	Strategic seasonal or inter-seasonal planning decisions to anchor the forecast in the plan for the season or for the year	Tactical intra-seasonal (on-farm) decisions to capitalize on the opportunities created by the forecasts	Operational intra-seasonal (on-farm) decisions to capitalize on the opportunities created by the forecasts	Emergency action decisions or contingency measures to avoid likelihood of negative impacts associated with extreme or severe weather events
Seasonal and inter-annual forecast	Decisions on what to plan for and what to plan with, such as on crop varieties to plant, land preparation methods, etc.			
Monthly (extended range) forecast		Decisions on when to do what, such as on on-farm activities – weeding, <u>pest</u> and disease monitoring,		
<u>Dekadal</u> (medium range) forecast			Decisions such as on on-farm activities – weeding, pest, and disease monitoring,	
Weekly (short-range) forecast			Decisions such as on agrochemical application, fertilizer application, crop harvesting, crop air-drying time, etc.	
Daily (short-range) forecast				Decisions such as on agrochemical application, fertilizer application, crop harvesting, crop air-drying time, etc.

In the process of developing the tool described in this paper, similarities and differences between traditional/community and scientific forecast were identified and described. Some of them are presented here. Similarities of indigenous and conventional forecasts were found to be as follows: -

- Both are based on observation of nature
- Both have a scientific basis
- Both can forecast onset and the quality of the season to come
- Both are accompanied by warnings on likely risks and hazards
- Both use signs/indicators and patterns
- Both use observance of wind
- Both can forecast below normal rainfall or an impending dry spell

Differences between traditional/community and scientific forecast as discussed are presented in Table 4

Table 4: Similarities and differences between indigenous and conventional forecasts

Attribute	Indigenous traditional method	Conventional scientific method
1. Amount of rainfall	• Can only guess amounts, e.g., more, less, etc.	• Able to present amount (e.g., 500mm)
2. Scenarios (terminologies used)	• Much, Little, Moderate, Average	• Normal, Above Normal, Below Normal
3. Distribution (Space/locality, Time (days, months))	• May not be able to guess	• Poorly estimated but can be estimated (in time and space)
4. Presentation	• Mostly oral	• Mainly visual
5. Prediction	• Uses local historical trends to predict future projections • Can forecast from 1 day up to 2 months or 3 months after indicator is seen	• Uses global historical trends to predict future projections • Can forecast up to and beyond 3 months
6. Onset	• Can make a guess of onset	• Can calculate an estimate of onset
7. Cessation	• Generally, not capable of guessing cessation but some indicators are emerging which can give hints in the middle of a season	• Can calculate an estimate of cessation
8. Duration of rain	• Can guess that the rains will be longer	• Can calculate start and end date
9. Length of season	• Can guess length of season (by intensity of frog croaking)	• Can calculate length of season
10. Off-season rains	• Difficult to deal with off-season rains	• Can give a clear indication of off-season rains
11. Nature of forecast	• Narrow in coverage and only momentary in relevance • Culture-based and may be interpreted differently for different areas • Not always documented in ways that farmers can understand • Does not apply weather instruments and probability • Cannot forecast amounts, <u>distribution</u> and end date • Process/ procedures are not documented • Emphasizing relationships between multiple parameters/ indicators • Easily available and accessible for use in agriculture by locals • Culture-based and interpreted differently for different areas • Easy to interpret and easy to make decisions based on the observations made • Relies mostly on holistic assessment of parameters/ indicators to inform decisions	• Has a wide spatial-temporal coverage/relevance • Not culture-based and is interpreted uniformly for different areas • Not always documented in ways that farmers can understand • Applies weather instruments and probability • Can forecast amounts, <u>distribution</u> and end date • Process/procedures well-documented • Focusing on individual parameters/ indicators of high predictive value • Not easily available and accessible for use in agriculture by locals • Not always documented in ways that farmers can understand • Difficult to interpret and not easy to make decisions based on the probabilistic information given • Relies mainly on individual mean values of parameters/ indicators to formulate advisories
12. Potential extent of extreme and/or severe weather events	• Not able to quantify the potential	• Able to quantify the potential

2.8 Key Conclusions

The tool provides opportunity for systematic analysis and empirical comparison of ITK-LTK weather forecasting with conventional weather forecasting as opposed to subjective anecdotal comparison that currently goes on in co-production initiatives and mystical, ritualistic rainmaker models. Based on the weather information generated, the TILKIT indicator assessment facilitates the determination of local lead times for scalable strategic, tactical, operational and emergency decisions in local agro-weather risk management, additional to conventional weather forecast, which often comes late. The approach adopts a diverse user participation that varies considerably in scope, ranging from individuals to user groups to communities, as opposed to the usual deferment to mystical, ritualistic “rainmaker” model which is often gender-biased and socially exclusive. The tool was found robust enough to be replicable, scalable and good for institutionalization of ITK-LTK weather forecasting under local contexts. The tool is considered a living instrument that can be used to share traditional forecast information that are updated regularly as per local arrangements – e.g., pre-season, onset of season, monthly, weekly, daily. The efficacy of TILKIT is premised on monitoring and observing as many local indicators as possible, tabulating the observations and their implications, and summing up their predictions to obtain forecast probabilities. Depending on the number of TILIs a community identifies and the number of signs each TILI exhibits, it is possible to tally the observations and estimate percentages of the more, less or normal rainfall predictions (normal, above or below normal) in the community.

2.9 Key Recommendations

To apply the tool at scale, it will need to be institutionalized and embedded in frontline extension frameworks to strengthen self-sustaining community climate information networks, with support from the authorities for the data networks, media and private sector. The tool can be applied earlier in the season before the conventional weather forecast is released, to serve as community preparedness mechanism for the season. For the tool to be successfully and effectively integrated with the conventional climate and weather forecast information, the conventional forecasting will need to be downscaled to the level of the traditional forecasting – the community level. Evidence from this work has been used to influence policy thinking and sensitization for decision-making at CEMIRIDE. The tool can be used for sensitization to influence policy reforms and decision-making at various levels among relevant stakeholders in the region.

Session 3: Side Events

The conference also had Side Events on the following topical areas: -

1. Climate information services in climate-smart agriculture: enhancing resilient agrifood
2. The contributions of agroecology to climate change
3. Annual Meeting of the Continental Think-Tank for responding to Emerging issues in African agriculture
4. Gaining Grounds...a workshop to strengthen collaborations to advance climate smart agriculture in Africa
5. Soil Initiative for Africa (SIA) Stakeholders Consultation (see available recordings)

Session 4: Results/ Outputs/ Outcomes

The CSA component of the SPAC 2022 conference was concluded with a Durban+10 Interim Synthesis Report at the ACSAF Plenary. The Report, a professional review presented by Dr Genevesi Ogiogio of the African Centre for Institutional Development (ACID), covered Africa's policy actions, lessons and recommendations in the implementation of Agriculture Sector decisions of the COPs from COP17 (2011, Durban) and COP26 (2021, Glasgow). The closing session was also blessed with a presentation on Interim Decadal Africa CSA Implementation Plan (CSAIP 2022-2032). The CSA options in the CSAIP 2022-2032 integrates provisions for relevant, locality-specific traditional and innovative practices, technologies and services that are socially sustainable and economically viable and environmentally friendly. The options include country readiness assessment survey and a review of agriculture sector and food systems adaptation and mitigation measures in the NDCs, and the development and update of CSA policy, strategy, investment plan, financing arrangements, capacity building programmes and support systems. The CSAIP 2022-2032 also recommends the development of incentive systems for CSA practices and innovation at all levels.

The 2011-2021 Synthesis Report synthesizes the extent to which agriculture and food systems have featured in the UNFCCC-COP deliberations and decisions since COP17 brought agriculture into the negotiations. It also assesses policies and programs that resulted from the COPs and the extent to which the policies and programs influenced CSA and food systems. The analysis provided insights and input for the Africa Group of Negotiators (AGN) and was intended to inform Africa's position at COP27 in 2022 and future COPs. Its findings are also meant to shape the push for a more structured support for agriculture and food systems in adaptation and mitigation financing by climate investment funds.

The report finds that: -

1. COP17 brought agriculture into the negotiations
2. COP22 launched the Adaptation of African Agriculture (AAA)
3. COP23 launched KJWA and mandated the SBSTA and SBI to jointly consider agriculture
4. Other UNFCCC-originated policies & programmes relevant to agriculture & food systems are NDCs and NDC-aligned NDPs, NAPAs, KJWA, AAAP
5. AU Green Recovery Action Plan (GRAP 2021-2027)
6. AU climate change & resilient development strategy & action plan 2022-2032
7. Several climate finance initiatives from multilateral finance organizations to the agriculture sector in Africa

The report concludes that, though fair to say that policy and program responses have been enhanced especially since NDCs, COP decisions on agriculture have so far not had a significant effect on policy responses to agriculture and food systems issues. The progress made in policy response is not commensurate with the flow of funds to agriculture and food systems.

The report recommends, among others: -

1. More effort and more concrete steps on KJWA
2. Enhanced funding for agriculture and food systems
3. Request for a new UNFCCC SB on agriculture and food systems
4. Explore Carbon Credit for debts owed by African countries. The repayments should go into the Africa Climate Change Fund (ACCF)
5. Proposal to establish a support system to compensate smallholder farmers for ecosystem services and for transition to a more climate-resilient agriculture
6. Issue CAADP green bonds for more climate-resilient food system, and create an African Credit Guarantee Facility to support the green bonds and other climate financing instruments
- 7.

Session 5: Next Steps

Recommendations of the conference to be continued at COP27.

Session 6: Closing Remarks

Closing remarks were delivered in Plenary <https://www.youtube.com/watch?v=ILkniU56xsg> , including an ACSAF Plenary https://faraafrica.zoom.us/rec/share/9zOzhcj2uwQR1mL7QQPJJa-klhKeAhqyf7fXo22En_HL8wbSKYE9B3CGRsaZgeh5a.MedXXfiYG2uqrx7b, by various speakers.

Annexes

Annex 1: Conference Programme

Conference Program Dashboard

Day 1: 14th September 2022

Time	Program Content								
	Joint Plenary Session / Zoom Link https://faraafrica.zoom.us/j/87449603891?pwd=YnduRTAwN0YwNHdpZDRJQUptIT2krUT09 Meeting ID: 87449603891; Passcode: 266211								
0900 to 1015	Opening Remark Dr. Alioune Fall Chair of FARA	Goodwill message Dr. Hans Hoogeveen GACSA co-chair	Goodwill message ADG- FAO Dr Abebe Haile-Gabriel	Video message Stephen Quest DG JRC European Union Commission	Goodwill message Mariya Gabriel , European Commissioner for Innovation, Research, Culture, Education and Youth	Goodwill message Massimo Mina , Head of Cooperation, Delegation of the European Union to Ghana	Video message HE Josefa Sacko Commissioner of DARBE, AUC	Video message Dr. Akin Adesina President of Africa Development Bank	Welcome Address Dr. Owusu Afriyie Akoto Minister of Food and Agriculture, Ghana
	Keynote Address HE. Jeanine Milly Cooper Minister of Agriculture, Liberia								
1015 to 1045	Coffee Break / Group Photograph/Networking								
1045 to 1130	Lead Paper “Introspection on CSA Action to Strengthen Accountability, Resource Use and Impact in Africa” Dr. Bruce Campbell (ClimEAT)								

	CSA Biennial Conference Thematic Paper Presentation						LEAP4FNSSA	PANAP Pan African Network for Economic Analysis Partners Meeting and launch of Reports	KM4AgD Knowledge Management for Agricultural Development Conference 2022
	Theme 1	Theme 2	Theme 3	Theme 4	Theme 5	Theme 6			
1130 to 1330	The role of digitalization in advancing CSA in the smallholders' systems Facilitator: Dr. Vincent Aduramigba [GCA] Venue: Britannia	The nexus of CSA and the mechanization of the smallholder system in Africa. Facilitator: Dr. Dr Mkomwa [ACTN] Venue: Volta Hall	Bottom-up and system-wide capacity development approaches to enhance CSA practices Facilitator: Prof. Babaloia Venue: Afram Hall	Strengthening the support of extension and advisory services to ensure CSA compliance among smallholder farmers in Africa Facilitator: Max Olupot [AFAAS] Venue: Abban Lounge	Advances in CSA technology generation and use in the crop, livestock, fisheries, and aquaculture. Facilitator: Dr Alcade Segnon [AICCRA] Venue: Kaizer Hall	Compatibility assessment of agroecology and CSA practices Facilitator: Dr. Cliff Dlamini [DeSIRA-Lift] Venue: Ridge Arena			
	Green House : Volta hall								
1330 to 1430	Lunch								
1430 to 1530	Theme 1 [Cont.]	Theme 2 [Cont.]	Theme 3 [Cont.]	Theme 4 [Cont.]	Theme 5 [Cont.]	Theme 6 [Cont.]			
1530 to 1700	ACSAF Plenary 1: Facilitator Dr. Manyewu Mutamba (AUDA-NEPAD)- RIDGE ARENA Commissioned study Foresight: Plausible futures of CSA in Africa Dr. Nasiru Taura Commissioned study Strategies for Continuous development of CSA technologies in Africa Prof. Tunde Ajayi								
1800	SPAC Cocktail: Poolside								

Day 2: 15th September 2022

Time	Side Events					LEAP4FNSSA WRITE SHOP Day II 	Partners Meeting Day II Asante Hall 	Conference Day II Prince Hall 
8:30 – 10:00 am	“ Climate information services in climate-smart agriculture: enhancing resilient agrifood”  Venue: Abban Lounge	The contributions of agroecology to climate change DeSIRA LIFT  Venue: Royal Suite	Annual Meeting of the Continental Think-Tank for responding to Emerging issues in African agriculture Venue: Afram Hall 	Gaining Grounds...a workshop to strengthen collaborations to advance climate-smart agriculture in Africa  Venue: Kaizer	Soil Initiative for Africa (SIA) Stakeholders Consultation  Venue: Ridge Arena			
10:30 – 17:00	Field Visit Take off @12:00  Field Group 1 Blue Skies Company Limited.							
Funders meeting Britannia	Writeshop 1 Botsio hall	Writeshop 2 Britannia	Writeshop 3 Abban Lounge	Writeshop 4 Royal Suite	Writeshop 5 Afram hall	Writeshop 6 Volta hall		

Time	ACSAP Plenary 2: @RIDGE ARENA- Session Chair: Dr. Alioune Fall	LEAP4FNSSA	Partners meeting Day III	Conference Day III
	Commissioned study 10 years analysis of Africa's progress on climate change / Decadal Plan for ACSAP Dr. Ogiogio Technical panel Discussion [Wole Fatunbi] Towards a common Africa position at CoP27 and COP28. Dr. Vanlauwe [CGIAR]; Dr. Akinbamijo [FARA]; Dr. Chapoto [ReNAPRI]; Dr. Litha Magingxa [ARC-South Africa]; Open Discussions	GA Day III Britannia 	Asante hall 	Prince hall 
0800 to 1030				
1030 to 1110	Coffee break			
1100 to 1330	Joint Closing Plenary: RIDGE ARENA			
	Project Pitch			
	Science: Biennial CSA			
	Policies: PANAP Statement			
	Knowledge Management			
	Partnerships: International Research Consortium (IRC) Launch			
	Closing Lunch			

Detailed Program

Joint Opening Plenary

14th September 2022

Venue: Ridge Arena [Main Conference Hall]

Time		Program Content	
Joint Plenary Session [0900 to 1100]			
Zoom Link: https://faraafrica.zoom.us/j/87449603891?pwd=YnduRTAwNHQyYwNHdpZDBJQUUpITZlcUtoQ9 Meeting ID: 87449603891 Passcode: 266211			
Opening Courtesies ** Ghana National Anthem ** Africa Union Anthem		MC: Julius Bizimungu (CNBC) Ghana Police Band	
	3 Minutes	Curtain Raiser [Video]	
	5 minutes	Welcome message Dr. Alioune Fall Chair of FARA	
	5 minutes	Goodwill message [Virtual] Dr. Hans Hoogeveen GACSA co-chair	
	5 minutes	Goodwill message Dr Abebe Haile-Gabriel ADG- FAO	
	5 minutes	Video message Stephen Quest DG JRC European Union Commission	
	5 minutes	Mariya Gabriel, European Commissioner for Innovation, Research, Culture, Education and Youth Goodwill message	
	5 minutes	Massimo Mina, Head of Cooperation, Delegation of the European Union to Ghana	
	5 minutes	Goodwill message [Video Message] HE Josefa Sacko Commissioner of DARBE, AUC	
	5 minutes	Goodwill message Dr. Akin Adesina [Video Message] President of Africa Development Bank	
	10 minutes	Opening Address Dr. Owusu Afriyie Akoto Minister of Food and Agriculture, Ghana	
20 minutes	Keynote Address HE. Jeanine Milly Cooper Minister of Agriculture, Liberia		
1100 to 1130		Coffee Break / Group Photograph/Networking	
30 minutes	Lead Paper "Introspection on CSA Action to Strengthen Accountability, Resource Use and Impact in Africa" Dr. Bruce Campbell (ClimEAT)		
Close of Opening Plenary			

2nd Plenary

16th September 2022

Venue: Ridge Arena [Main Conference Hall]

Time: Plenary

MC: Julius Bizimungu (CNBC)

Zoom <https://faraafrica.zoom.us/j/87449603891?pwd=YnduRTAwNOYwNHdpZDRJQUpiT2krUT09>

Meeting ID: 87449603891

Passcode: 266211

Time	Conference	Time	Activity particulars
1100 to 1220	Partners Report & Pitch	5 min	Traditional Dance / MC Introductions
		5 min	Reporting on side events [Dr. Wale Olayide]
		20 min	Partners organizations and projects pitch [MC]
	Science Biennial CSA	10 min	Summary of Papers presentations Best Paper Award
		10 min	Book Introduction: Strategies for continuous development of CSA technologies [Launching: Dr. Alioune Fall]
		10 min	Outcomes of ACSAF Discussions / Policy implications
	Knowledge management	10 min	Conference Communique /Awards/ Certificates
		10 min	Induction of 2022 Fellows into the KM4AgD CoP and Center of Excellence
		10 min	Unveiling of the Good Practice Report on partnerships
	Policies: PANAP	15 min	PANAP Meeting Report Irchad Razaaly, Head of Delegation of the European Union to Ghana /EU Ambassador to Ghana (present physically)

5 min	Traditional Music and Dance
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PARTNERSHIPS Launch of the IRC		Founding Launch of The International Research Consortium (IRC) Co-moderated by Dr Kwesi Attah-Krah and Dr Shadrack Moephuli
Time (GMT)	Topic	Responsible
	Launch Context and Initial Remarks	
12.20- 12.25	Remarks (AU and EU)	Video message from Co-Chairs of the HLPD WG on FNSSA:

		Ms. Magdalena Gajdzinska, [Policy Officer, EC DG Research and Innovation] Dr. Guillermo Cardon, [Policy Officer EC DG Agriculture, and Rural Development] Dr. Monica Ebele Idinoba, [Co-Chair of the HLPD WG on FNSSA]
	Presentation of the International Research Consortium (IRC)	
12.30- 12.45	Presentation of the Summary of the IRC	Dr. Irene Annor-Frempong [Coordinator, LEAP4FNSSA, FARA] •
	Goodwill Messages	
12.45-13.05	Goodwill messages from founding members	Dr Christof Weissteiner, [LEAP4FNSSA Project Officer, EC REA] Member States [Ghana, Senegal, South Africa, Kenya, Egypt, France] Research institutions and Regional Bodies [FARA, CAADP-XP4, CIRAD, WUR, RUFORUM, AGRINATURA, GFAR, CGIAR, YPARD, etc]
	Official Launch of the IRC	
13.05 – 13.15	Outlining the Principles of the FNSSA priority on which the IRC Communique / Declaration is based	Nienke Buisman, [EC DG RTD, International Cooperation Policy] Dr Hans-Joerg Lutzeyer, [EC DG RTD]
13.15- 13.20	Reading of the IRC Declaration	Dr, Aggrey Agumya, [Director of Research and Innovation, FARA]
13.20-13.30	Official Launch of the IRC	Honorable Dr. Kwaku Afriyie [Minister of Environment Science and Technology (MESTI), Ghana] Irchad Razaaly, [Head of Delegation of the European Union to Ghana /EU Ambassador to Ghana (present physically)]
		Photography

	Closing Remarks: ED of FARA
1330	Lunch



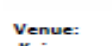
Detail Program

Day 1: CSA Biennial CSA Conference

Thematic Paper Presentation						
	Theme 1	Theme 2	Theme 3	Theme 4	Theme 5	Theme 6
1130 to 1330	The role of digitalization in advancing CSA in the smallholders' systems Facilitator: Dr. Vincent Aduramigba [IAR&T] Venue: Britannia	The nexus of CSA and the mechanization of the smallholder system in Africa. Facilitator: Dr. Dr Mkomwa [ACTN] Venue: Volta Hall	Bottom-up and system-wide capacity development approaches; to enhance CSA practices Facilitator: Prof. Babalola Venue: Afram Hall	Strengthening the support of extension and advisory services to ensure CSA compliance among smallholder farmers in Africa Facilitator: Max Oluopot [AFAAS] Venue: Abban Lounge	Advances in CSA technology generation and use in the crop, livestock, fisheries, and aquaculture. Facilitator: Dr Alcade Segnon [AICCRA] Venue: Kaizer Hall	Compatibility assessment of agroecology and CSA practices Facilitator: Isolina Boto [DeSIRA-Lift] Venue: Ridge Arena
1330 to 1430	Lunch					
1430 to 1530	Theme 1 [Cont.]	Theme 2 [Cont.]	Theme 3 [Cont.]	Theme 4 [Cont.]	Theme 5 [Cont.]	Theme 6 [Cont.]
1530 to 1700	ACSAP Plenary 1: Facilitator Dr. Manyewu Mutamba (AUDA-NEPAD) Commissioned study Foresight: Plausible futures of CSA in Africa Dr. Nasiru Taura Commissioned study Strategies for Continuous development of CSA technologies in Africa Prof. Tunde Ajayi					
1800	SPAC Cocktail: Poolside					

Day 2: CSA Biennial CSA Conference

Day 2: 15th September 2022

Time	Side Events					WRITE SHOP Day II	Partners Meeting Day II	Conference Day II
8:30 – 10:00 am	“Linking climate information services to climate-smart agriculture practices: optimizing CSA across multiple scales”  Venue: Abben Lounge	The contributions of agroecology to climate change  Venue: Royal Suite	Annual Meeting of the Continental Think-Tank for responding to Emerging issues in African agriculture  Venue: Afram Hall	Gaining Grounds...a workshop to strengthen collaborations to advance climate-smart agriculture in Africa  Venue: Kaizer	Soil Initiative for Africa (SIA) Stakeholders Consultation  Venue: Ridge Arena			
10:30 – 17:00	Field Visit							
	Take off @ 12:00 	Field Group 1 Blue Skies Company Limited.						

Day 3: CSA Biennial CSA Conference

Time	ACSAP Plenary 2: Session Chair: Dr. Alioune Fall @ Ridge Arena
0830 to 1030	Commissioned study 10 years analysis of Africa's progress on climate change / Decadal Plan for ACSAP Dr. Ogiogio Technical panel Discussion [Wole Fatunbi] Towards a common Africa position at CoP27 and COP28. Dr. Vantauwe [CGIAR]; Dr. Akinbami [FARA]; Dr. Chapoto [ReNAPRI]; Dr. Litha Magingxa [ARC-South Africa]; Open Discussions
1030 to 1100	Coffee break
1100 to 1400	Joint Plenary & Closing Lunch

Annex 2: Conference Participants for Sub-Thematic Area 3

Sub-theme 3: Bottom-up and system-wide capacity development approaches to enhance CSA practices

Facilitator: Prof. Babalola // **Venue:** Afram Hall

Name of presenter	Paper Title	Abstract No
1. Anthony	Evaluating Climate-Smart Adaptation Practices on Cocoa Insect Pests and Diseases Incidences Among Farmers in Cross River State, Nigeria	25
2. Agbongiarhuoyi		
3. Emily Nguyen	Advancing a Gender-Responsive Delivery Model for Mitigating Climate Impacts in Ghana's Cocoa Landscape	26
4. Perperidis		
5. Joshua Sikhu Okonya	The role of CSA Multistakeholder platforms in advancing the CSA agenda in Eastern and Southern Africa Physical	27
6. Lydiah C Miriti	Factors Influencing Access to Rural Finance Market by Different Actors in Climate-Smart Agriculture in Kenya	28

7. Michael Omodara	Deployment of monitoring devices to address quality degradation in bagged grain in storage	29
8. Moses Odeke	Scaling CSA initiatives in Eastern and Central Africa: Successes and lessons learned	30
9. Mtisunge Mngoli	A Comparative Analysis of The Soyabean Value Chain in Malawi and Zambia	31
10. Munoko M. Karen	Capacity Assessment within Multi-Stakeholder Partnerships Towards Resilient Food Systems: Imperatives for Climate Smart Agriculture	32
11. Nelson C.K. Sello	Climate-Smart Agriculture Practices: Interrogating the CSA Stakeholder engagement in Botswana	33
12. Olabisi Damilola 13. Omodara	Factors influencing choice of climate change adaption practices among cassava farmers and its effect on cassava productivity in Osun State, Nigeria	34
14. Oladele Idowu	Mainstreaming Climate-Smart Agriculture into Agricultural Extension Curricula in West Africa: The Roles of Sasakawa Africa Association	35
15. Olga Mapanje	Financing sustainable agriculture in Sub-Saharan Africa: the role of FinTech	36
16. Osumba, Joab	Towards a replicable innovative community based ethnographic tool for adaptive local weather forecasting using indigenous or traditional indicators to strengthen local agro weather resilience at scale	37
17. Temitope Oluwaseun Ojo	Financing adoption of climate smart agriculture among smallholder rice farmers in Osun State, Nigeria	38
18. Akyala Abraham	Using Gender-Land Rights to Increase Climate Resilience Among Smallholder Farmers in the least Developed Countries (LCDS)	39
19. Aliku Orevaoghene	Yield Response of Sorghum to Micronutrient Fortified Fertilizer in the Savanna Agroecological Zone of Nigeria	40
20. Diriba Tulu	Adaptation Capacity of Indigenous Sheep Breed to Saline Lake Drinking Water in Dry Area of Ethiopia Under Climate Change Scenarios	41
21. Ermias Kebreab	Empowering African Scientists in development of livestock nutrient requirements and greenhouse gas emissions tracking system	42
22. Sunita Facknath	Climate-Smart Agriculture as an innovative socioeconomic sector for the national economy	43
23. Zakaria Fouad Fawzy	Applied of Climate Smart Agriculture Approach for Agricultural Development to African Food Security and Sustainability of Agriculture as well as Adaptation Future Climate Changes	44
24. Philippa Ojimelukwe	Climate adaptation strategy for cassava processing and storage by a Niger Delta Community in Nigeria	45

Annex 3: Conference convenors and partners

Conference partners with FARA included the following organisations and agencies: -



Biennial Africa
Climate Smart Agriculture
Stakeholders Conference



ACCRA, GHANA



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